CSE4203 Principles of Software Architecture

RapidMed

Emergency Triage and Real-Time Ambulance Dispatch System

Milestone 1: Proposal and architectural requirements

Task 1: System overview

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1. System overview

The problem it solves

An ambulance crew with a seriously ill or injured patient needs to know which emergency departments can take the patient now and how long each will take to reach, and the receiving team needs to know what is coming. Without a shared, live view of capacity, the choice rests on distance and phone calls.

RapidMed gives the crew, the hospital and the dispatch centre the same live picture. On the ambulance's computer, the paramedic records vital signs and a triage level in an app that works without a signal, and opens the patient's emergency summary from Sanad, the patient health data management system (PHDMS). RapidMed ranks nearby hospitals by free beds of the type needed and by traffic-aware travel time, reserves a bed at the chosen one and sends the patient's details ahead, while a dispatch supervisor watches the fleet and can override any routing decision.

Who uses it

User and the part they use	What they do
Field paramedic Ambulance computer app	Records vitals and triage level, even offline; opens the patient's summary (break-glass if needed); gets the destination, route and alerts
Hospital triage coordinator Web portal, hospital view	Keeps bed availability up to date; accepts incoming patients and prepares the team using their vital signs and arrival time
Central dispatch supervisor Web portal, dispatch view	Monitors ambulances and hospital capacity on a live map; overrides routing; acts on alerts, such as a crew losing its connection

Why it is architecturally interesting

1. **Consistency against availability.** A bed can go to only one patient, so reservations must be strictly consistent: two crews must never be sent to the last free resuscitation bay. Yet crews work on roads with weak or no signal and must keep recording and reading data offline. When the network is partitioned, a system cannot guarantee both, so the design must decide which actions may go ahead offline and which must wait for the server.
2. **Security and privacy against usability and speed.** Health data are sensitive personal data under Egypt's data protection law, and patient histories need encryption, role-based access and an audit trail. But an unconscious patient cannot consent, and the crew cannot wait for approval: the record must open within seconds through a "break-glass" path that overrides normal access control while recording who used it and why. Caching records on the ambulance's computer helps speed and offline work but leaves sensitive data on a device shared by several crews.
3. **Dependence on outside services.** Routing, patient summaries and notifications rely on three external services, any of which can slow down, fail or hit a usage limit. None of these failures may stop a crew from getting a destination, so the design needs time-outs, fallbacks and circuit breakers, and a mock of each service to test failures.